

IMAGINARY ELECTRIC COMPANY

Power Generation Division - Plant Integration

Alpilla 600 MW Combined Cycle Power Plant - 1 x IE-9H.02 multi-shaft power island

Generators, GCB and Isolated Phase Busduct - Datasheet

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Addressee	Istanbul EPC Muhendislik ve Taahhut A.S. (consortium leader) - EPC engineering
Contract	ALP-EPC-001 (Owner contract); consortium agreement IEPC-IEC

	Prepared	Checked	Approved
Function	IEC plant integration engineer	IEC lead engineer (discipline)	IEC project engineering manager
Signature / date	signed 2026-12-04	signed 2026-12-04	signed 2026-12-04

Rev	Date	Description
0	2026-09-27	Issued for EPC engineering (reference data)
0	2026-09-27	Issued for EPC engineering
A	2026-12-04	TQ-IEC-025: tags of neutral earthing / excitation transformers and protection panels

CASE STUDY - Imaginary Electric, its products and all data are fictional.

1. Generator data

Item	GT generator IE-GH560	ST generator IE-GA270
Rated apparent power	560 MVA	270 MVA
Rated power factor / active power	0.85 lagging / 476 MW	0.85 lagging / 229.5 MW
Reactive capability at rated active power	0.85 lagging to 0.95 leading (ER-04.08)	0.85 lagging to 0.95 leading
Rated voltage / frequency / speed	21.0 kV +/- 5 % / 50 Hz / 3,000 rpm	15.75 kV +/- 5 % / 50 Hz / 3,000 rpm
Cooling	hydrogen-cooled rotor and stator core, 4.0 barg H ₂ ; water-cooled stator winding	TEWAC (air, water coolers)
Efficiency at SRC load (incl. excitation)	98.90 %	98.80 %
Losses to cooling water at SRC	4.3 MW (H ₂ coolers + stator water)	2.2 MW (air coolers)
Excitation	static, from generator terminals via excitation transformer	static
Short-circuit ratio / Xd" (unsaturated)	0.50 / 0.21 pu	0.52 / 0.18 pu
Insulation / temperature rise	class F / class B	class F / class B
Generator circuit breaker (GCB)	included, IEC 62271-37-013, 21 kV, 20 kA rated current, 130 kA	included, 15.75 kV, 12.5 kA, 100 kA
Isolated phase busduct (IPB)	generator - GCB - GSU LV, with tap-off to unit auxiliary transformer (UAT by EPC)	as GT
Stator transport weight / dimensions	340 t / 10.8 x 4.6 x 4.9 m	185 t / 8.6 x 4.1 x 4.2 m
Rotor weight (maintenance lift)	72 t	46 t
H ₂ / CO ₂ consumption	H ₂ make-up 12 Nm ³ /day; CO ₂ for purging 150 Nm ³ per purge	-
Protection	IEC generator protection panels (IEEE C37.102 functions, duplicated main 1 / main 2)	as GT

2. Interfaces

- H₂ and CO₂ bottle racks and supply piping to the IEC gas control panel: **by EPC** (IEC-ALP-DOR-001 item 5.6).
- Closed cooling water for H₂ coolers, stator water coolers and TEWAC coolers: by EPC (IEC-ALP-REQ-001 table 5.1).
- UAT tap-off from the IPB between GCB and GSU: IPB tap-off flange by IEC, UAT and MV busduct by EPC.
- Neutral earthing: high-resistance earthing transformer in the generator neutral cubicle by IEC.

3. Tagged items (Rev A, TQ-IEC-025)

Tag	Item	Rating
10MKA20	GT generator neutral earthing transformer with secondary resistor	21 kV / 240 V, 50 kVA, 10 s; limits earth fault current to 8 A
20MKA20	ST generator neutral earthing transformer with secondary resistor	15.75 kV / 240 V, 40 kVA, 10 s; 7 A
10MKC30	GT excitation transformer	21 / 0.9 kV, 5.5 MVA, dry type
20MKC20	ST excitation transformer	15.75 / 0.7 kV, 2.5 MVA, dry type
10MKA30	GT generator protection panels (main 1 / main 2)	IEC 61850, 220 V DC
20MKA30	ST generator protection panels (main 1 / main 2)	IEC 61850, 220 V DC